

Problem Set 2

THE CHALLENGES OF GROWTH

Problem 1 Carbon Emissions and Growth. It is possible to extend the production function so that output is produced with labor input, N ; capital input, K , and carbon-intensive energy, E . If we write a production function as

$$Y = N^{1/3}K^{1/3}E^{1/3}$$

- 1) Using the input values below, does this production function exhibit constant returns to scale?

Units of N	Units of K	Units of E	Output
27	125	64	
54	250	128	
81	375	192	

- 2) Solve for the necessary amount of energy to produce the specified level of output in each row below.

Units of N	Units of K	Units of E	Output
54	250		120
54	375		120
81	250		120

- 3) Comment on the following statement using your results from 2).

"Because of constant returns to scale, in order for output to rise, we must increase our consumption of energy."

- 4) Use words to describe how output can increase without an increase in the use of energy. The text points out that there is widespread agreement among economists that a tax on carbon emissions energy is the best way to reduce carbon emissions. How does this proposition fit into the production function above?

FINANCIAL MARKETS AND EXPECTATIONS

Problem 1 Approximating the Price of Long-Term Bonds. *The present value of an infinite stream of dollar payments of $\$z$ (that starts next year) is $\$z/i$ when the nominal interest rate, i , is constant. This formula gives the price of a consol—a bond paying a fixed nominal payment each year, forever: It is also a good approximation for the present discounted value of a stream of constant payments over long but not infinite periods, as long as i is constant. Let's examine how close the approximation is.*

- Suppose that $i = 10\%$. Let $\$z = 100$. What is the present value of the consol?
- If $i = 10\%$, what is the expected present discounted value of a bond that pays $\$z$ over the next 10 years? 20 years? 30 years? 60 years? (*Hint: Use the formula from the chapter but remember to adjust for the first payment.*)
- Repeat the calculation in 1) and 2) for $i = 2\%$ and $i = 5\%$.

Problem 2 Monetary Policy and the Stock Market. Assume the short term real policy rate, current and expected, had been 2% until now. Suppose the Fed decides to tighten monetary policy and increase the short-term policy rate (r_{1t}) from 2% to 3%.

- 1) What happens to stock prices if the change in r_{1t} is expected to be temporary, that is, last for only one period? Assume that expected real dividends do not change. Use equation (14.17).

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots$$

- 2) What happens to stock prices if the change in r_{1t} is expected to be permanent, that is, is expected to persist? Assume that expected real dividends do not change. Use equation (14.17).
- 3) What happens to stock prices today if the increase in the real interest rate, current and expected, reflects an increase in expected future output and expected future dividends?

EXPECTATIONS, CONSUMPTION, AND INVESTMENT

Problem 1 Individual Saving and Aggregate Capital Accumulation. Suppose that every consumer is born with zero financial wealth and lives for three periods: youth, middle age, and old age. Consumers work in the first two periods and retire in the last one. Their income is \$5 in the first period, \$25 in the second, and \$0 in the last one. Inflation and expected inflation are equal to zero, and so is the real interest rate.

- 1) What is the present discounted value of labor income at the beginning of the first period of life? What is the highest sustainable level of consumption such that consumption is equal in all three periods?
- 2) For each age group, what is the amount of saving that allows consumers to maintain the constant level consumption you found in 1)? (*Hint: Saving can be a negative number if the consumer needs to borrow to maintain a certain level of consumption.*)
- 3) Suppose there are n people born each period. What is total saving in the economy? (*Hint: Add up the saving of each age group. Remember that some age groups may have negative saving.*) Explain.
- 4) What is total financial wealth in the economy? (*Hint: Compute the financial wealth of people at the beginning of the first period of life, the beginning of the second life, and the beginning of the third period. Add the three numbers. Remember that people can be in debt, so financial wealth can be negative.*)

Problem 2 Borrowing Constraints and Aggregate Capital Accumulation. Continue with the setup from Problem 1, but suppose now that borrowing constraint do not allow young consumers to borrow. If we call the sum of income and total financial wealth "*cash on hand*", then the borrowing restriction means that consumers cannot consume more than their cash on hand. In each age group, consumers compute their total wealth and then determine their desired level of consumption as the highest level that allows their consumption to be equal in all three periods. However, if at any time, desired consumption exceeds cash on hand, then consumers are constrained to consume exactly their cash on hand.

- 1) Calculate consumption in each period of life. Compare this answer to 1) of Problem 1 and explain any differences.
- 2) Calculate total saving for the economy. Compare this answer to your answer to 3) of Problem 1 and explain any differences.
- 3) Derive total financial wealth for the economy. Compare this answer to your answer to 4) of Problem 1 and explain any differences.
- 4) Consider the following statement: "Financial liberalization may be good for individual consumers, but it is bad for overall capital accumulation." Discuss.

EXPECTATIONS, OUTPUT, AND POLICY

Problem 1 The New *IS-LM*. For each of the changes in expectations in 1) through 4), determine whether there is a shift in the *IS* curve, the *LM* curve, both curves, or neither. In each case assume that no other exogenous variable is changing.

- 1) a decrease in the expected future real interest rate
- 2) an increase in the current real policy interest rate
- 3) an increase in expected future taxes
- 4) a decrease in expected future income

Problem 2 Effects of Tax Cuts. A new president, who promised during the campaign that she would cut taxes, has just been elected. People trust that she will keep her promise, but expect that the tax cuts will be implemented only in the future. Determine the impact of the election victory on current output, the current interest rate, and current private spending under each of the assumptions in 1) through 3). In each case, indicate what you think will happen to Y^e , r^e , and T^e , and then how these changes in expectations affect output today.

- 1) The Fed will not change its current policy rate, but does not make a commitment as to the direction of rates in the future.
- 2) The Fed will act to prevent any change in current and future output.
- 3) The Fed will not change either then current real policy interest rate or the future real policy interest rate.

THE OPEN ECONOMY

Problem 1 Balance of Payments. Consider two fictional economies, one called the domestic country and the other the foreign country. Given the transactions listed in 1) through 7), construct the balance of payments for each country. If necessary, include a statistical discrepancy.

- 1) The domestic country purchase \$100 in oil from the foreign country.
- 2) Foreign tourists spent \$25 on domestic ski slopes.
- 3) Foreign investors were paid \$15 in dividends from their holdings of domestic equities.
- 4) Domestic residents gave \$25 to foreign charities.
- 5) Domestic businesses borrowed \$65 from foreign banks.
- 6) Foreign investors purchased \$15 of domestic government bonds.
- 7) Domestic investors sold \$50 of their holdings of foreign government bonds.

Problem 2 Uncovered Interest Parity. Consider two bonds, one issued in euros (€) in Germany, and one issued in dollars (\$) in the United States. Assume that both government securities are one-year bonds—paying the face value of the bond one year from now. The exchange rate, E , stands at 0.75 euros per dollar.

The face values and prices on the two bonds are given by

	Face Value	Price
United State	\$10,000	\$9,615.38
Germany	€10,000	€9,433.96

- 1) Compute the nominal interest rate on each of the bonds.
- 2) Compute the expected exchange rate next year consistent with uncovered interest parity.
- 3) If you expect the dollar to depreciate relative to the euro, which bond should you buy?

- 4) Assume that you are a US investor and you exchange dollars for euros and purchase the German bond today. One year from now, it turns out that the exchange rate, E , is actually 0.72 (0.72 euros buys one dollar). What is your realized rate of return in dollars compared to the realized rate of return you would have made had you held the US bond?
- 5) Are the differences in rates of return in 4) consistent with the uncovered interest parity condition? Why or why not?

Problem 3 The Goods Market in an Open Economy. Consider an open economy characterized by the following equations:

$$\begin{aligned} C &= c_0 + c_1(Y - T) \\ I &= d_0 + d_1Y \\ X &= x_1Y^* \\ IM &= m_1Y \end{aligned}$$

The parameters x_1 and m_1 are the propensities to export and import, respectively. Assume that the real exchange rate is fixed at a value of 1 and treat foreign income, Y^* , as fixed. Also assume that taxes are fixed and that government purchases are exogenous (i.e., decided by the government). We explore the effectiveness of changes in G under alternative assumptions about the propensity to import.

- 1) Write the equilibrium condition in the market for domestic goods and solve for Y .
- 2) Suppose government purchases increase by one unit. What is the effect on output? Assume that $0 < m_1 < c_1 + d_1 < 1$. Explain why.
- 3) How do net exports change when government purchases increase by one unit?

Now consider two economies, one with $m_1 = 0.5$ and the other with $m_1 = 0.1$. Each country is characterized by $(c_1 + d_1) = 0.6$.

- 4) Suppose one of the economies is much larger than the other. Which economy do you expect to have the larger value of m_1 ? Explain.
- 5) Calculate your answers to 2) and 3) for each economy by substituting the appropriate parameter values.
- 6) In which economy will fiscal policy have a larger effect on output? In which economy will fiscal policy have a larger effect on net exports?

Problem 4 Exchange Rate Regimes. An overvalued real exchange rate is a rate such that domestic goods are too expensive relative to foreign goods, net exports are too small, and by implication the demand for domestic goods is too low. This leads to difficult policy choices for the government and central bank. The equations (* denotes a value for the foreign country) that describe the economy are:

The IS curve:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right)$$

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The Phillips curve for the domestic and the foreign economy:

$$\text{Domestic Phillips curve : } \pi - \bar{\pi} = \frac{\alpha}{L}(Y - Y_n)$$

$$\text{Foreign Phillips curve : } \pi^* - \bar{\pi}^* = \frac{\alpha^*}{L^*}(Y^* - Y_n^*)$$

- 1) We assume that the foreign economy is always in medium-run equilibrium. What are the implications of that assumption for foreign output and foreign inflation?

- 2) We assume that the domestic and foreign economies share the same anchored value for the level of expected inflation denoted so that $\bar{\pi} = \bar{\pi}^*$. What is the implication of that assumption once both the domestic and foreign economies are both in medium-run equilibrium?
- 3) Draw the $IS - LM - UIP$ diagram for the case where the domestic country has an overvalued nominal exchange rate. What is the key feature of that diagram? Under fixed exchange rates without a devaluation, how does the economy return to its medium-run equilibrium?
- 4) Draw the $IS - LM - UIP$ diagram for the case where the domestic country has an overvalued nominal exchange rate. Show how the economy can return to its medium-run equilibrium when a devaluation is a policy choice.
- 5) Recall that the assumption has been made that interest rate parity holds, so $i = i^*$ at all times. Compare the returns on the domestic bond and the returns on the foreign bond in the period of the devaluation. Will bond holders continue to believe there is a completely fixed nominal exchange rate? If bond holders believe that another devaluation is possible, what are the consequences for domestic interest rates?

BACK TO POLICY

Problem 1 Deficit Reduction as a Prisoner's Dilemma Game. Suppose there is a budget deficit. It can be reduced by cutting military spending, by cutting welfare programs, or by cutting both. The Democrats have to decide whether to support cuts in welfare programs. The Republicans have to decide whether to support cuts in military spending. The possible outcomes are represented in the following table:

(A game with a payoff structure like the one in this problem, and that produces the outcome, is known as a prisoner's dilemma.)

		Welfare Cuts	
		Yes	No
Defense Cuts	Yes	(R=1,D=-2)	(R=-2,D=3)
	No	(R=3,D=-2)	(R=-1,D=-1)

The table presents payoffs to each party under the various outcomes. Think of a payoff as a measure of happiness for a given party under a given outcome. If Democrats vote for welfare cuts, and Republicans vote against cuts in military spending, the Republicans receive a payoff of 3, and the Democrats receive a payoff of -2 .

- 1) If the Republicans decide to cut military spending, what is the best response of the Democrats? Given this response, what is the payoff for the Republicans?
- 2) If the Republicans decided not to cut military spending, what is the best response of the Democrats? Given this response, what is the payoff for the Republicans?
- 3) What will the Republicans do? What will the Democrats do? Will the budget deficit be reduced? Why or why not? Is there a way to improve the outcome?